

The Zero-Sum Game of Mutual Fund Marketing

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Competitive Equilibrium

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Mutual Fund Marketing: What We Know

- Asset management has grown dramatically — AUM and fund counts have surged over three decades; **investor search costs** are a first-order friction (Hortaçsu & Syverson, 2004; Huang, Wei & Yan, 2007)
- U.S. mutual funds spend roughly **one-third of fee revenue on marketing**; advertising, broker distribution, and visibility drive significant fund-level inflows (Sirri & Tufano, 1998; Gallaher, Kaniel & Starks, 2008)
- A growing literature studies **social media marketing** of funds in China: De Haan et al. (2025), Kacperczyk & Pagnotta (2026), and Li et al. (2026) examine mutual fund livestreaming on retail platforms

An open question

Prior work evaluates marketing at the **fund level**. But marketing decisions are made at the **family level**. Does fund-level marketing translate into family-level growth?

The Missing Family-Level Test

- Marketing **expands the family** only if inflows represent **new money** from outside the family
- If marketing primarily **reallocates existing assets** within the family, then fund-level effects do not translate into family growth
- Most prior work evaluates marketing at the **fund level**, even though marketing decisions are made — and success should be measured — at the **family level** (Gallagher et al., 2008)

This paper:

1. Evaluate livestreaming at both fund *and* family level using **25,685 events** on China's leading platform
2. Decompose within-family capital flows to identify reallocation vs. new attraction
3. Analyze **795 transcripts** with an LLM to understand which communication strategies successfully attract outside capital

Setting and Data

Institutional Setting: Mutual Fund Livestreaming

A large-scale novel marketing channel:

- From 2020, COVID-19 disrupted traditional sales; fund families adopted **real-time broadcasts** on retail platforms
- **Tiantian Fund** — leading third-party platform — hosts the broadcasts
- AMAC formalized regulatory guidance in 2021

Key institutional features:

- Production is **family-level**: dedicated staff, family channel
- Sharp selection: < 4 funds featured per broadcast from 25+ products
- **Shopping cart**: viewers purchase featured funds in one click

Identification gift

Within the same family and same quarter, some funds are featured while others are not.

This within-family, within-quarter variation identifies both the direct effect on featured funds *and* the spillover to non-featured siblings.

The Tiantian Fund Livestreaming Platform

(1) Platform Portal



(2) Livestream Room



(3) Shopping Cart



Data and Sample

Livestream data (Tiantian Fund):

- **25,685** livestreaming events, July 2020 – June 2025
- Treatment: fund in shopping cart = “featured”

Fund universe:

- **5,809** actively managed equity/hybrid funds
- **180** fund families
- **90,850** fund-quarter observations
- Exclude: index, closed-end, QDII, umbrella funds

Descriptives:

- Mean flow: 0.38% SD: 42%

Flow measurement:

Fund flow (return-adjusted):

$$\text{Flow}_{i,q} = \frac{\text{TNA}_{i,q} - \text{TNA}_{i,q-1}(1 + r_{i,q})}{\text{TNA}_{i,q-1}}$$

Family flow: AUM-weighted aggregate of fund flows within family f :

$$\text{FamilyFlow}_{f,q} = \frac{\sum_{i \in f} [\text{TNA}_{i,q} - \text{TNA}_{i,q-1}(1 + r_{i,q})]}{\sum_{i \in f} \text{TNA}_{i,q-1}}$$

Captures whether livestreaming expands the family's investor base or merely redistributes assets.

Empirical Design

Two estimating equations:

Family-level TWFE – does the family grow?

$$\text{FamilyFlow}_{f,q} = \alpha_f + \delta_q + \beta_1 \text{Livestream}_{f,q} + \mathbf{X}'_{f,q-1}\Gamma + \varepsilon_{f,q}$$

Fund-level TWFE – who gains, who loses?

$$\text{FundFlow}_{i,q} = \alpha_i + \delta_q + \beta_1 \underbrace{\text{FundLivestream}_{i,q}}_{\text{fund } i \text{ is featured}} + \beta_2 \underbrace{\text{FamilyLivestream(Oth)}_{i,q}}_{\text{sibling fund is featured}} + \mathbf{X}'_{i,q-1}\Gamma + \varepsilon_{i,q}$$

Fixed effects:

- Fund / family FE absorb time-invariant unobservables
- Year-quarter FE control for aggregate conditions

Robustness:

- Family $\times$ quarter FE: absorbs all time-varying family shocks
- SE clustered at family level throughout

Main Result: Zero-Sum at the Family Level

Family-Level Flows: Null Across All Specifications

Eight family-level TWFE specifications:

Variable	Coef.	(SE)
Family First Livestream	3.77	(5.12)
Family Post Livestream	-0.66	(2.34)
Family Livestream	0.90	(2.17)
Family Market Share	-15.67	(18.43)
Market Livestream (Oth)	-0.90	(1.65)
Family Livestream Count	-0.09	(0.50)

Family FE, quarter FE, controls. SE clustered at family level.
None approach conventional significance.

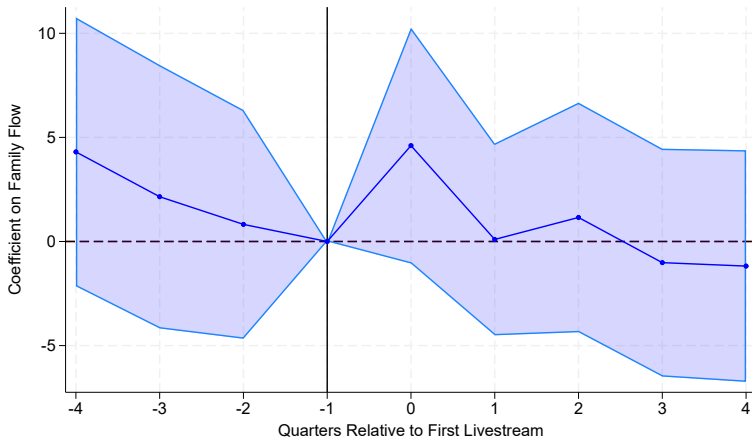
Our finding

Livestreaming generates **zero** change in family-level flows across all eight specifications. Confidence intervals rule out economically meaningful effects.

What is happening?

If inflows to featured funds are real, they must come from *within* the family – at the expense of non-featured siblings. We test this directly.

Event Study: No Family-Level Response Around Adoption



TWFE estimates of family flows around initial livestream adoption. Pre-event coefficients are flat and insignificant (parallel trends). No sustained increase post-adoption.

Fund-Level Decomposition: The Reallocation Mechanism

	(1)	(2)	(3)	(4)	(5)
Fund First Livestream	14.13*** (1.89)			12.95*** (1.96)	
Fund Livestream			14.87*** (1.85)		13.27*** (1.93)
Family Livestream (Oth)				-1.49** (0.70)	-1.94** (0.88)
Controls	No	No	No	No	Yes

Fund FE, quarter FE. SE clustered at family. ***, **, * = 1%, 5%, 10%.

Featured funds

Gain **+13 to +15 pp** in quarterly flows. $\approx$ one-third of within-fund flow SD.

Non-featured siblings

Lose **-2 pp** — opposing sign directly establishes within-family cannibalization.

Robustness: Family $\times$ Quarter Fixed Effects

Replace fund + quarter FE with family $\times$ quarter FE:

- Absorbs *all* time-varying family characteristics: aggregate family flow, family performance, market conditions
- Mechanically forces AUM-weighted within-family flows to sum to the family's aggregate flow
- Since family aggregate = 0 (Table 1), the coefficient on featured funds = **full reallocation magnitude**

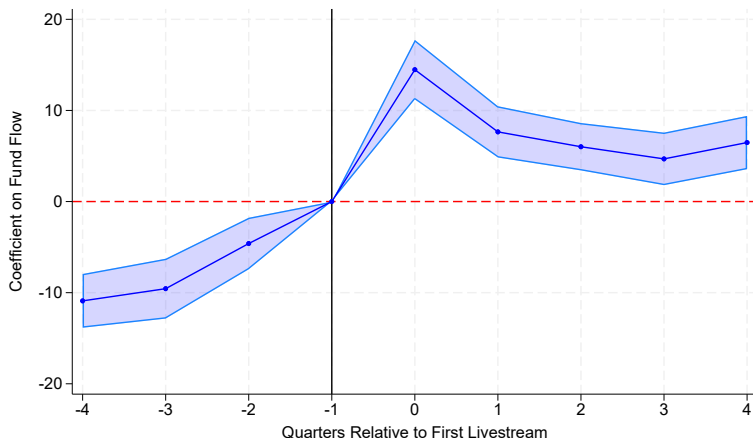
	TWFE	Family $\times$ Quarter FE
Fund Livestream	14.87*** (1.85)	17.59*** (2.25)

Interpretation

The **+17.6 pp** under family $\times$ quarter FE represents the dollar-for-dollar reallocation: every dollar flowing into featured funds is offset by an equal dollar flowing out of non-featured siblings

Why Livestream? Performance Dynamics

Performance of Featured vs. Non-Featured Funds



Monthly performance of featured funds vs. non-featured siblings relative to the livestream event month. Shaded bands: 95% CI. Fund and month fixed effects.

Interpretation: Families as Open Market Operators

Pattern before the livestream:

- Featured funds significantly **outperform** siblings in months preceding the event
- Non-featured funds underperform $\Rightarrow$ they are *overcapitalized* (Berk & Green, 2004)

Pattern after the livestream:

- Inflows into featured funds erode their outperformance
- Outflows from non-featured funds improve their performance
- Performance gap **compresses** – consistent with flow-performance equilibrium

Mechanism: Internal open market operation (Berk, 2017)

Families have superior information about which funds are undercapitalized. Livestreaming **steers existing investors** from overcapitalized to undercapitalized funds – without attracting new outside capital.

Benefit: Retains investors who might otherwise search outside the family.

Fee Reallocation and Revenue: Also Zero

Could livestreaming shift flows toward higher-fee products?

- **Management fees** (majority of revenue): **no response** in any specification or horizon
- **Service fees**: modest temporary increase of a few basis points
- **Loads**: slight subsequent reduction
- **Flow-induced revenue**: null across all specifications and horizons

Variable	Contemp.			One-quarter ahead		
	(1)	(2)	(3)	(4)	(5)	(6)
Livestream coef.	−0.01 (0.01)	−0.01 (0.01)	0.01 (0.01)	−0.01 (0.01)	−0.01 (0.01)	0.00 (0.01)

Dep. var.: flow-induced revenue scaled by beginning family assets.

Bottom line

Families extract **no additional revenue** from livestreaming. The zero-sum reallocation does not improve profitability.

Which Livestreams Actually Attract Outside Capital?

Question: While most livestreaming is intra-family reallocation, some families do attract new outside money. What distinguishes them?

Approach:

- Define *abnormal family flows* = residuals from regression of family flows on livestream intensity + controls
- Sort into High (top 5%) vs. Low (bottom 5%) within each quarterly cross-section
- Randomly sample 5 transcripts per family $\Rightarrow$ **795 transcripts**

LLM scoring (Qwen-2.5-32B):

- Score each transcript on **16 dimensions**: sentiment, emotional intensity, sales pressure, past performance, social proof, educational content, risk transparency, community building, ...
- Each scored 1–10 using a *fixed rubric defined ex ante*
- Segment-level (10k chars), averaged to transcript level

Results: Only “Hard Information” Differentiates

Mean differences: High vs. Low abnormal-flow families

Dimension	Difference	p-value
Past Performance Focus	+0.28	0.02
Social Proof Reliance	+0.23	0.04
Sentiment Positivity	+0.08	0.51
Emotional Intensity	−0.05	0.68
Sales Pressure	+0.11	0.39
Educational Content	+0.03	0.82
Risk Transparency	−0.09	0.44
Community Building	+0.06	0.59
8 further dimensions: all insignificant		

2 of 16 dimensions matter

Successful families emphasize:

- **Past performance** — hard evidence of recent returns
- **Social proof** — third-party recognitions, peer adoption

Implication

Investors respond to **verifiable information** consistent with rational learning (Berk & Green, 2004).

Emotional appeals, presentation style, and engagement tactics do *not* move capital.

Competitive Pressure: Why Families Must Livestream

If own-livestreaming generates no family-level gains, why do families persist?

Within families (own vs. sibling):

- Own-fund livestreaming: +7.2 pp flows
- Sibling livestreaming: -4.9 pp flows
- Net family effect: ≈ 0

Across families (market competition):

- When rivals increase livestreaming, non-livestreamers lose ≈ 27 pp flows per 1-SD increase in market activity
- Own market share predicts +620 pp flows per unit increase

Prisoner's dilemma

Families must livestream **not to attract new investors** but to **defend existing assets** from rivals who are more actively marketing.

Unilateral exit from livestreaming results in immediate flow losses.

Industry is trapped in a wasteful promotional arms race (Roussanov, Russ & Wei, 2021).

Conclusion

Summary

- (1) Zero-sum at the family level.** Livestreaming generates large fund-level inflows (+14 pp) but **zero** family-level flow or revenue change. Featured funds gain at the direct expense of non-featured siblings.
- (2) Intra-family capital optimization.** Families promote recently outperforming (undercapitalized) funds. Post-event performance converges — consistent with Berk-Green rational learning. Livestreaming is an **internal open market operation**.
- (3) Defensive arms race.** Families livestream to **defend market share**, not to expand it. Market-wide competition creates a prisoner's dilemma: exit means flow losses, stay means zero net gain.
- (4) Hard information wins.** The rare cases of successful inter-family capital attraction rely exclusively on **past performance and social proof** — not sentiment or engagement.

Broader Implications

- **For the mutual fund literature:** Fund-level marketing effects $\nrightarrow$ family growth. Flow-performance sensitivities must be evaluated at the family level.
- **For investor welfare:** Livestreaming facilitates learning and improves within-family capital allocation — but aggregate marketing may be socially excessive given the competitive arms race with no net new capital formation.
- **For policy:** AMAC's 2021 guidance correctly targets investor education and compliance. Our evidence confirms livestreaming can be informative (hard-info channel) but also has a defensive, rent-defending motive.

Thank you. Questions welcome.

Appendix: Full Family-Level Flow Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Family First Livestream	3.77 (5.12)							
Family Post Livestream		-0.66 (2.34)						
Family Livestream			0.90 (2.17)				1.07 (2.19)	
Family Market Share				-15.67 (18.43)				
Market Livestream (Oth)					-0.90 (1.65)			
Family Livestream Count						-0.09 (0.50)		0.01 (0.50)
Market-Family Count							24.55 (20.33)	14.52 (18.47)
Family FE	Y	Y	Y	Y	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y

Appendix: Fund-Level Decomposition (Full)

	(1)	(2)	(3)	(4)	(5)
Fund First Livestream	14.13*** (1.89)			12.95*** (1.96)	
Fund Post Livestream		14.64*** (2.01)			
Fund Livestream			14.87*** (1.85)		13.27*** (1.93)
Family Livestream (Oth)				-1.49** (0.70)	-1.94** (0.88)
Fund FE	Y	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y	Y
Controls	No	No	No	No	Yes

***, **, * = 1%, 5%, 10%. SE clustered at family level.

Appendix: Share Class Analysis

Specification: Fund-quarter FE absorb all fund-level variation.

$$\text{ClassFlow}_{c,q} = \delta_{f(c),q} + \beta_1 \text{ClassLivestream}_{c,q} + \mathbf{X}'_{c,q-1}\Gamma + \varepsilon_{c,q}$$

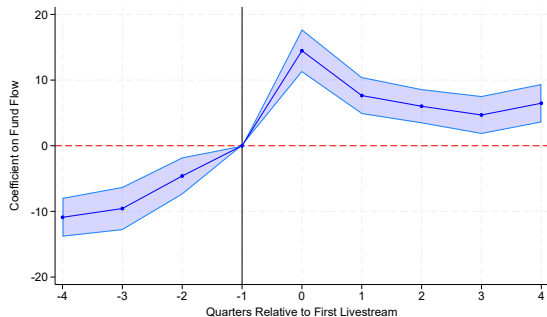
	(1)	(2)
Class Livestream	10.20*** (2.34)	
Solo Livestreamer (non-feat. class)		-9.82 (8.15)
Has Sibling Livestreamer (non-feat. class)		-19.91 (14.60)

Fund $\times$ quarter FE. SE clustered at family level.

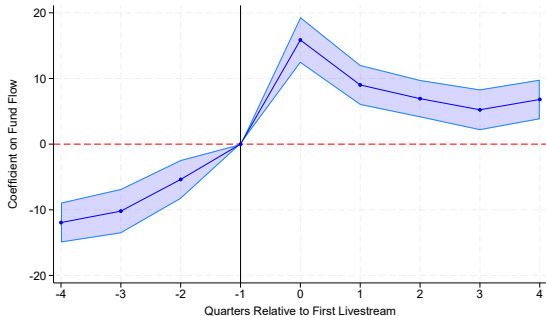
Featured classes gain **+10 pp** relative to non-featured classes of the *same fund* in the same quarter. Reallocation operates at every level: family $\rightarrow$ fund $\rightarrow$ share class.

Appendix: Performance Event Study (Robustness)

TWFE



Family $\times$ Quarter FE



Both specifications confirm pre-event outperformance of featured funds and post-event convergence. Results are robust to controlling for all time-varying family characteristics.

Appendix: LLM Scoring Dimensions (All 16)

1. Sentiment Positivity
2. Past Performance Focus*
3. Social Proof Reliance*
4. Emotional Intensity
5. Sales Pressure
6. Educational Content
7. Risk Transparency
8. Community Building
9. Urgency / FOMO
10. Product Complexity Framing
11. Presenter Credibility Signals
12. Interactive Engagement
13. Narrative / Storytelling
14. Regulatory Compliance Emphasis
15. Future Performance Claims
16. Comparative Benchmarking

* Significantly different between high and low abnormal-flow families ($p < 0.05$).

Scoring: Qwen-2.5-32B; 10,000-character overlapping segments; 1–10 scale; fixed rubric defined ex ante so measurement error is orthogonal to communication strategies under study.